



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

PH1005-1 – PHYSICS FOR BUILDINGS AND CONSTRUCTIONS
(CV)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

List of constants: Velocity of light, $c = 3 \times 10^8 \text{ ms}^{-1}$, Planck's constant, $h = 6.63 \times 10^{-34} \text{ Js}$, Electron mass, $m = 9.11 \times 10^{-31} \text{ kg}$, Electron charge, $e = 1.602 \times 10^{-19} \text{ C}$, Permittivity of vacuum, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$, Boltzmann constant, $k = 1.38 \times 10^{-23} \text{ J/K}$, Avogadro number, $N_A = 6.023 \times 10^{26} / \text{kg mole}$, Acceleration due to gravity, $g = 9.81 \text{ m/s}^2$

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- For a particle executing SHM along x-axis force is given by
 A) $-A kx$ B) $A \cos kx$
 C) $A e^{-kx}$ D) $A kx$
- The spring constant of a spring is K. When it is divided into n equal parts, then what is the spring constant of one piece?
 A) nK B) K/n
 C) k D) $(n+1)K/n$
- The time period of a simple pendulum is 2 seconds. If its length is increased by 4 times, then its period becomes
 A) 16 s B) 12 s
 C) 8 s D) 4 s
- When an external force is applied on an oscillating particle and reduces the amplitude of the oscillation of the particle then this is called as
 A) Simple harmonic motion B) Forced oscillation
 C) Standing wave D) Circular motion
- The ratio of lateral strain to linear strain within elastic limit for a material is
 A) Poisson ratio B) Young's modulus
 C) Rigidity modulus D) Bulk modulus
- In cantilever beams, there is _____ stress above neutral axis
 A) Compressive B) Temperature
 C) Tensile D) Shear
- If poisson's ratio of material is 0.2 and because of stress its change in length is 25% of original length, where as the change in its radius is %
 A) 5 B) 10
 C) 15 D) 25
- Modulus of rigidity is the ratio of
 A) Stress to strain B) Tensile stress to tensile strain
 C) Compressive stress to compressive strain D) Shear stress to shear strain
- Emission of a photon by an excited atom due to interaction with a passing photon nearby is called
 A) Spontaneous emission B) Induced absorption
 C) Stimulated emission D) Thermionic emission
- Metastable states are
 A) Ground state energy states. B) Excited state energy levels in which electrons stay for very short interval of time.
 C) Excited state energy levels in which electron can stay for unusually long time. D) Nuclear energy states.

11. Laser action is found in _____ semiconductor.
 A) Direct band gap B) Indirect band gap
 C) Germanium D) Silicon
12. In a semiconductor laser there are no mirrors used because
 A) The laser light required is of low power B) Mere driving a minimum current in the diode is enough for laser action
 C) P-section acts as one mirror and n-section acts as the other mirror D) A pair of parallel planes cleaved or polished at a particular angle in the crystal reflect the light efficiently
13. Which of the following is the basic principle of light signal communication through optical fibre?
 A) Total internal refraction of light B) Reflection of light
 C) Polarization of light D) Total internal reflection of light
14. Which of the following has more distortion?
 A) Single step-index fibre B) Graded index fibre
 C) Multimode step-index fibre D) Glass fibre
15. Attenuation means
 A) Loss of signal strength B) Division of signal strength
 C) Amplification of signal strength D) Tuning of signal
16. The numerical aperture of a fiber if the angle of acceptance in air is 15 degrees, is
 A) 0.17 B) 0.26
 C) 0.50 D) 0.75
17. Sound intensity is directly proportional to of wave amplitude.
 A) Square B) Square root
 C) Cube D) Magnitude
18. Speed of sound is maximum in
 A) Vacuum B) Gases
 C) Liquids D) Solids
19. Reverberation time is directly proportional to
 A) Effective surface area B) Volume of the room
 C) Lateral surface area D) Shape of the room
20. The average rate of transfer of sound energy per unit area perpendicular to the direction of propagation is known as of sound.
 A) Loudness B) Intensity
 C) Pressure D) Pitch

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) What is stiffness factor? Write its physical significance.	4	L*2	2	1,2
	b) With the help of the expression for amplitude of forced oscillations discuss the condition for resonance.	8	L2	1	1,2
	c) A particle executes simple harmonic motion of period 22 s and amplitude 0.1 m. Find the distance of the particle from the mean position in 3.5 s.	4	L3	1	1,2
2.	a) Write a note on sharpness of resonance.	4	L2	1	1,2
	b) What is Poisson's ratio? Write a note on limiting values of Poisson's ratio.	8	L2	2	1,2
	c) A load of 2 kg produces an extension of 1 mm in a wire of 2 m length and 1 mm in diameter. Find the Young's modulus of the material of the wire.	4	L3	2	1,2
3.	a) State Hooke's law. Draw stress strain diagram.	4	L2	2	1,2
	b) Derive an expression for Spring constant of Series and Parallel combination of springs.	8	L2	2	1,2
	c) A steel wire of radius 1 mm is bent into an arc of a circle of radius 50 cm. Calculate the bending moment. Given: Young's modulus is $20 \times 10^{10} \text{ Nm}^{-2}$.	4	L3	2	1,2

Unit – II

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|-------|---|---|----|---|-----|
| 4. a) | What is a laser? Explain the principle of spontaneous emission and stimulated emission of radiation. | 6 | L2 | 3 | 1,2 |
| b) | With necessary diagrams explain the construction and working of semiconductor laser. | 6 | L2 | 3 | 1,2 |
| c) | Calculate the wavelength of emission from a GaAs semiconductor laser whose band gap energy is 1.44 eV. | 4 | L3 | 3 | 1,2 |
| 5. a) | What is an optical fiber? Explain the principle on the basis of which optical transmission is achieved through a fiber. | 6 | L2 | 4 | 1,2 |
| b) | Derive an expression for numerical aperture of an optical fiber in terms of R.I. of core and cladding. | 6 | L2 | 4 | 1,2 |
| c) | The angle of acceptance of an optical fiber is 30° when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33 | 4 | L3 | 4 | 1,2 |
| 6. a) | What is attenuation in an optical fiber? Explain the possible reasons for the same. | 6 | L2 | 3 | 1,2 |
| b) | What are the basic components required for the construction of a laser? Explain the method of production of laser. | 6 | L2 | 4 | 1,2 |
| c) | The fractional refractive index of an optical fiber of diameter 80 μm is 0.03. The refractive index of the core is 1.48 and the operating wavelength is 0.8 μm . Calculate (i) acceptance angle (ii) critical angle for core-cladding interface and (iii) number of modes supported in the fiber. | 4 | L3 | 4 | 1,2 |

Unit – III

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|-------|--|---|----|---|-----|
| 7. a) | What is sound? Describe the classification of sound. | 6 | L2 | 5 | 1,2 |
| b) | Explain the requisites for acoustics in auditorium. | 6 | L2 | 5 | 1,2 |
| c) | Write a note on impact of Noise in Multi-storied buildings. | 4 | L2 | 5 | 1,2 |
| 8. a) | What is Weber-Fechner law? Explain. | 4 | L2 | 5 | 1,2 |
| b) | Derive Sabine's formula for reverberation time. | 8 | L2 | 5 | 1,2 |
| c) | Write a note on factors affecting the acoustics and remedial measures. | 4 | L3 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
